

Document Summary TASK-1A56581F

Analysis & Generated Response

Certainly! Below is an example of a Python code snippet for encryption using AES (Advanced Encryption Standard). This code includes necessary imports and a simple function to encrypt text.

```
```python
from Crypto.Cipher import AES
from Crypto.Util.Padding import pad, unpad

Define your secret key and initialization vector (IV)
key = b'Sixteen byte key'
iv = b'16byte iv'

def encrypt_text(text):
 # Create an AES cipher object with the specified key and IV
 cipher = AES.new(key, AES.MODE_CBC, iv)

 # Pad the text to be a multiple of 16 bytes (AES block size)
 padded_text = pad(text.encode(), AES.block_size)

 # Encrypt the padded text
 encrypted_data = cipher.encrypt(padded_text)

 return encrypted_data

Example usage:
original_text = "Hello, World!"
encrypted_data = encrypt_text(original_text)

print("Original Text:", original_text)
print("Encrypted Data:", encrypted_data.hex())
```
```

Explanation:

1. Imports:

- `from Crypto.Cipher import AES``: Imports the AES cipher module.
- `from Crypto.Util.Padding import pad, unpad``: Imports functions for padding and unpadding data.

2. Key and IV:

- The key is a 16-byte string (AES-128). You can replace this with any 16-byte value you choose.
- The IV is also a 16-byte string. It should be unique for each encryption operation to ensure security.

3. **Encryption Function:**

- ``encrypt_text(text)``: This function takes the original text as input.
- It creates an AES cipher object using the provided key and IV.
- The text is padded to a multiple of 16 bytes using ``pad()``.
- The padded text is then encrypted using ``cipher.encrypt()``.

4. **Example Usage:**

- An example message "Hello, World!" is converted to bytes and passed to the ``encrypt_text`` function.
- The original text and the encrypted data are printed.

Note:

- This code uses a simple padding scheme (PKCS7) for demonstration purposes. In practice, you might want to use more robust padding schemes like PKCS5 or ISO 10126.
- For production environments, consider using a secure key management system and storing the IV securely with